# PROTIK BISWAS



*Software Engineering* | *Quality Assurance* | *Applied AI Research*
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## PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 3.94/4.00; expected 2027) with hands-on experience in software development, QA automation and applied AI research. Developed PHP/MySQL systems, created a 32-case QA suite and automated workflows using Selenium and Pytest.

## CORE SKILLS

- **Programming:** Python, C++, Java, C#, JavaScript, PHP, SQL, R, MATLAB.
- **Software and QA:** Selenium WebDriver, Pytest, JUnit, test design, functional and regression testing, Git.
- **AI and data:** PyTorch, TensorFlow, scikit-learn, timm, Transformers, SciBERT, OpenCV, NumPy, pandas, SciPy.
- **Web and databases:** HTML, CSS, AJAX, JSON, WordPress, Figma, MySQL, Oracle, SQL Server, XAMPP.
- **Office and analytics:** Word, Excel, PowerPoint, SPSS.
- **Research:** experiment design, ablations, evaluation, Grad-CAM, technical writing.

## EXPERIENCE

**Software Intern at Trends Bird Limited**

- Translated requirements into responsive layouts and high-fidelity Figma prototypes.
- Incorporated design feedback and prepared developer handoff materials.

## EDUCATIONAL QUALIFICATIONS

**BSc in Computer Science |** American International University-Bangladesh (AIUB) | CGPA: 3.94/4.00 | Expected 2027

**Milestone School and College |** Higher Secondary Certificate and Secondary School Certificate, Science | GPA: 5.00/5.00

## SOFTWARE ENGINEERING AND QA PROJECTS

**Hostel Management System QA using Selenium, Pytest, PHP, MySQL:**

- Created 32 test cases, automated key workflows and validated MySQL records.

**Budget-Aware Mutation-Guided LLM Unit Testing using Qwen2.5-Coder, JUnit, Defects4J, Bugs.jar:**

- Compared baseline, repair, random and adaptive feedback under equal budgets with automated compile-and-run checks.

**Online Hostel Management System using PHP, MySQL, HTML, CSS, JavaScript:**

- Developed registration, room management, booking, payment and administration workflows.

## ADDITIONAL DEVELOPMENT AND DESIGN PROJECTS

- **Interactive WordPress Theme:** Figma and WordPress.
- **TechGeek:** Java application.
- **Gardening App:** Figma UI/UX prototype.
- **Animated Front-End Website:** HTML and CSS.
- **Supermarket Management System:** C# application.

- **Real-Time Heart Rate Detector:** Microprocessor prototype.
- **Scenario-Based Application:** C++ application.

## RESEARCH EXPERIENCE

- **DeepLabNet in Liver Ultrasound Segmentation:** Built a frequency- and boundary-aware DeepLabv3/MobileNetV3 model; achieved 0.8641 Dice and 0.7707 IoU with 11.023M parameters.
- **CerebriLite-CAM for Brain Tumor MRI Classification:** Built a MobileNetV2/CBAM four-class classifier with Grad-CAM; achieved 97.0% accuracy.
- **Rare Disease Onset Prediction using Orphadata:** Combined genetic, epidemiological, text and graph features using CatBoost, TF-IDF and Node2Vec.
- **LiteRCF-Net for Thyroid Ultrasound Classification:** Combined lesion, context and frequency branches; achieved 86.71% accuracy and 0.9065 AUROC on TN5000.

### RESEARCH PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- **CerebriLite-CAM: A Lightweight Interpretable Attention Network for Brain Tumor Classification from MRI** - accepted and presented at the IEEE 3rd International Conference on Computing, Applications and Systems (COMPAS 2026), Paper ID 634.
- **DeepLabNet: Lightweight Liver Ultrasound Segmentation via Haar Frequency Fusion and Entropy Guided Boundary Refinement** - accepted for presentation at IECBES 2026.
- **Predicting Rare Disease Onset Categories Using Integrated Genetic, Epidemiological, and Graph-Based Features from Orphadata** - accepted at the 4th International Conference on Computing Advancements.
- **Improved Deep PACFNet: Multimodal Deep Learning for Cardiac Anomaly Detection Using Synchronized ECG-PCG Signals** - accepted at IEEE SPICSCON 2026.
- **Vaccination Coverage and Challenges in Achieving Universal Immunization in Bangladesh** - received a certificate for this contribution at the 7th International Conference on Integrated Science.

## AWARDS AND ADDITIONAL INFORMATION

- **Awards:** Dean's List and merit-based bachelor's scholarship.
- **Certification and membership:** Cisco Networking certification; Engineering Students' Association of Bangladesh member.
- **Languages:** Bangla (native), English (fluent), Hindi (conversational).
- **Interests:** Research projects, UI/UX design, sketching, photography, gaming and virtual reality.

## REFERENCES

**Dr. Kamruddin Md. Nur**

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